# Bus Network Resilience in London and Los Angeles

## A Comparative Study of Bus Network Resillience Under Disruption

How do bus networks in London and Los Angeles respond to disruptions, and what insights can we gain about their resilience?

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# Networks used

- Compare the bus transportation networks of London and Los Angeles
  - Analyze the network and its resilience
- Networks are modeled as graphs where:
  - Nodes represent bus stops
  - Edges represent bus routes connecting consecutive stops

Filtered for bus stops only!

## London Bus Network

**NODES**: 21373 | **EDGES**: 27042
**Average Degree**: 2.53
**Average Clustering Coefficient**: 0.0159
**Global Clustering Coefficient**: 0.0313
**Average Path Length**: 30.32
**Diameter**: 97

*Source: https://transitfeeds.com/p/traveline/1033*

## Los Angeles Bus Network

**NODES**: 12007 | **EDGES**: 12957
**Average Degree**: 2.16
**Average Clustering Coefficient**: 0.0119
**Global Clustering Coefficient**: 0.0240
**Average Path Length**: 61.33
**Diameter**: 197

*Source: https://transitfeeds.com/p/la-metro/184*

# LONDON

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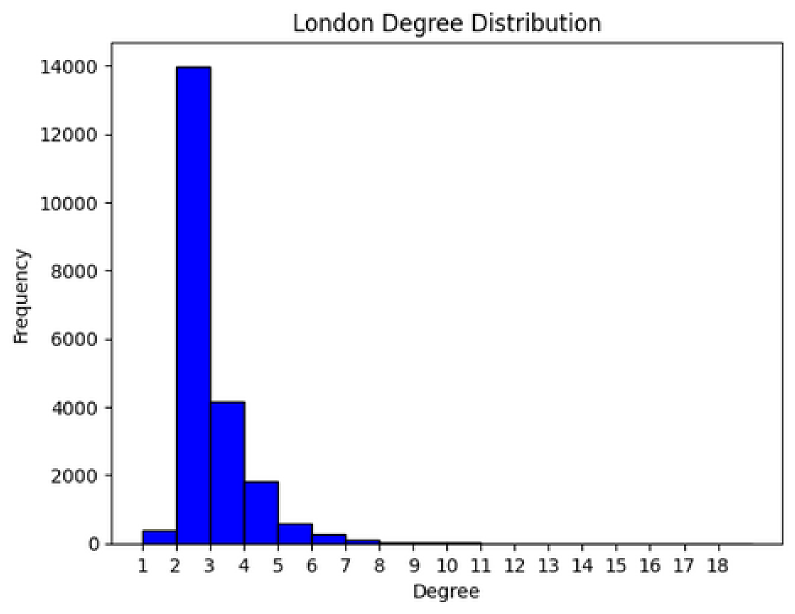
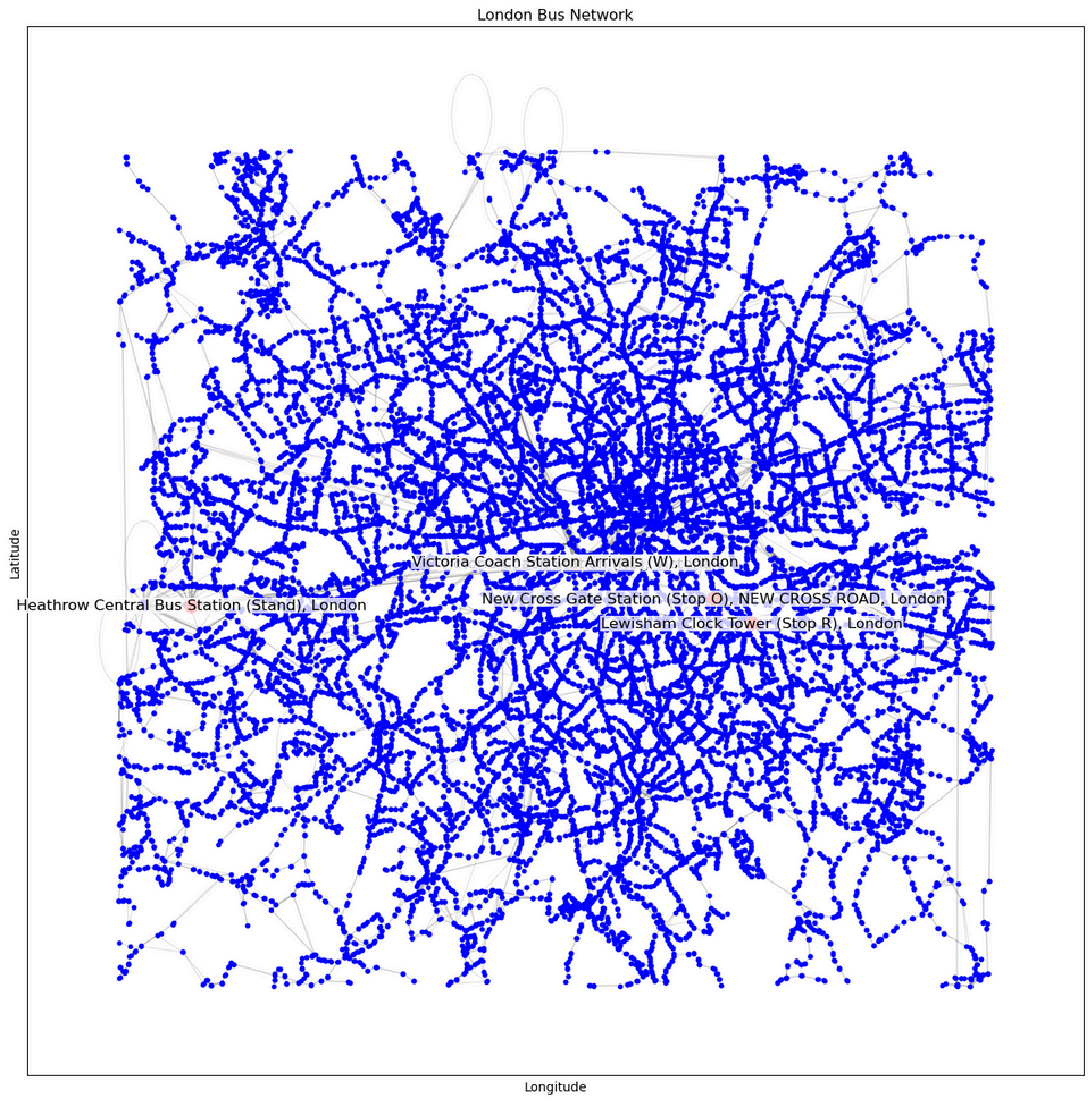



## Most Important Stations

| STATION NAME | BETWEENNESS CENTRALITY |
|---|---|
| Victoria Coach Station Arrivals (W) | 0.333009 |
| Victoria Coach Station (NE) | 0.317163 |
| Heathrow Central Bus Station (Stand) | 0.181231 |
| New Cross Gate Station (Stop O) | 0.162313 |
| Lewisham Clock Tower (Stop R) | 0.149888 |




*Note: "Victoria Coach Station (NE)" is right under "Victoria Coach Station Arrivals (W)"*

# LOS ANGELES

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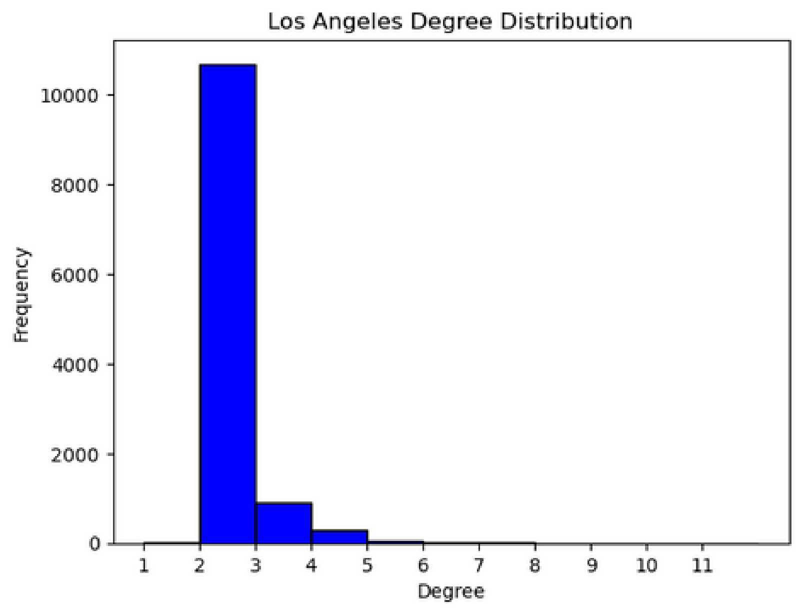
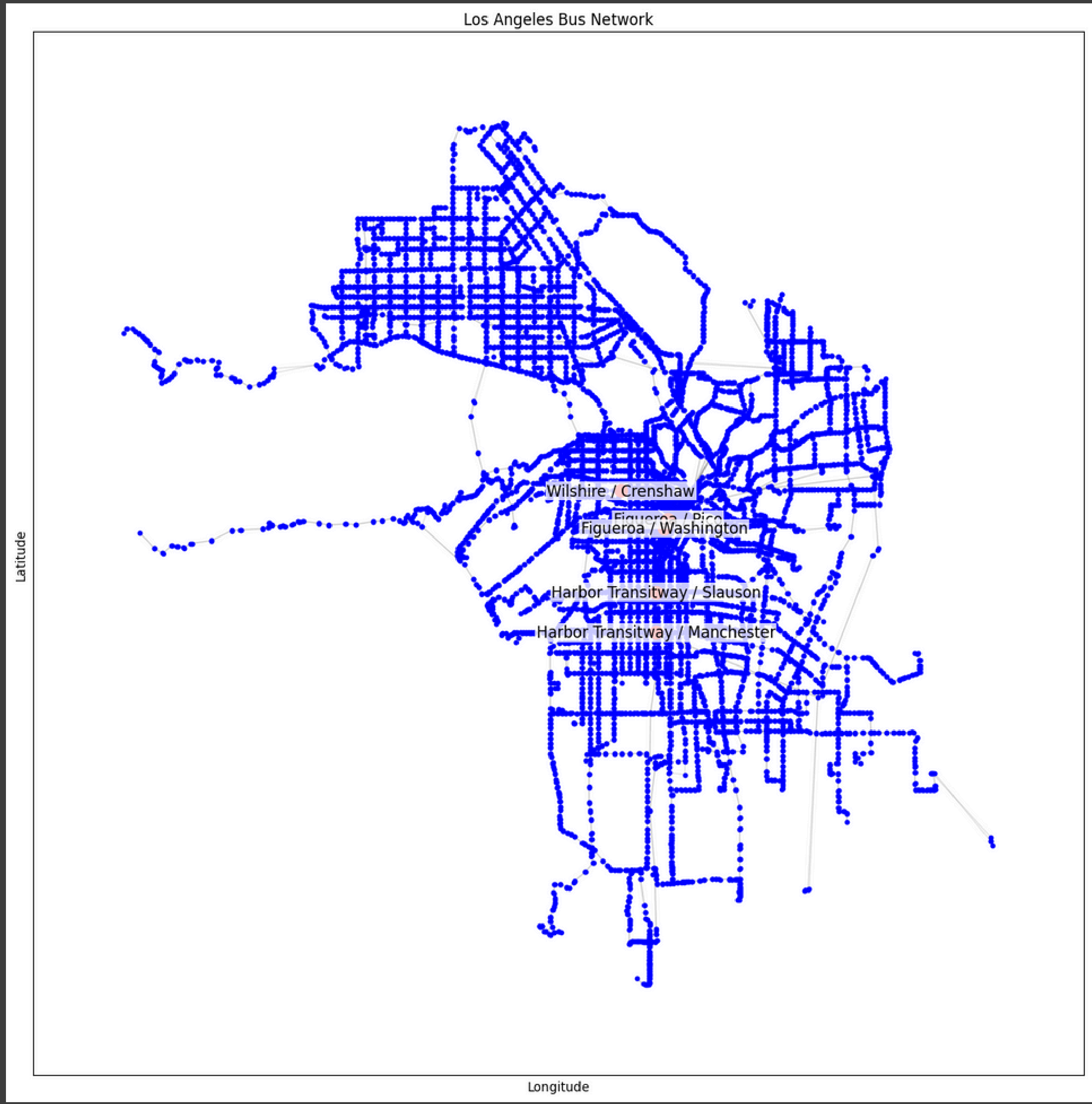



## Most Important Stations

| STATION NAME | BETWEENNESS CENTRALITY |
|---|---|
| Harbor Transitway / Slauson | 0.221865 |
| Figueroa / Pico | 0.214108 |
| Harbor Transitway / Manchester | 0.208528 |
| Figueroa / Washington | 0.201803 |
| Wilshire / Crenshaw | 0.199689 |



# Research Question

## How resilient are the London and Los Angeles bus networks to random and targeted failures?

### Point of Failure

- Point at which the network's performance experiences a significant breakdown due to node removal
- Threshold where the network transitions from functioning efficiently to being disrupted

## 1. Simulate <u>targeted</u> and <u>random</u> station removal

- Remove stations randomly to simulate unexpected disruptions
  - 3 repetitions and take the average of the results
- Remove stations based on betweenness centrality to simulate targeted disruptions on most important stations

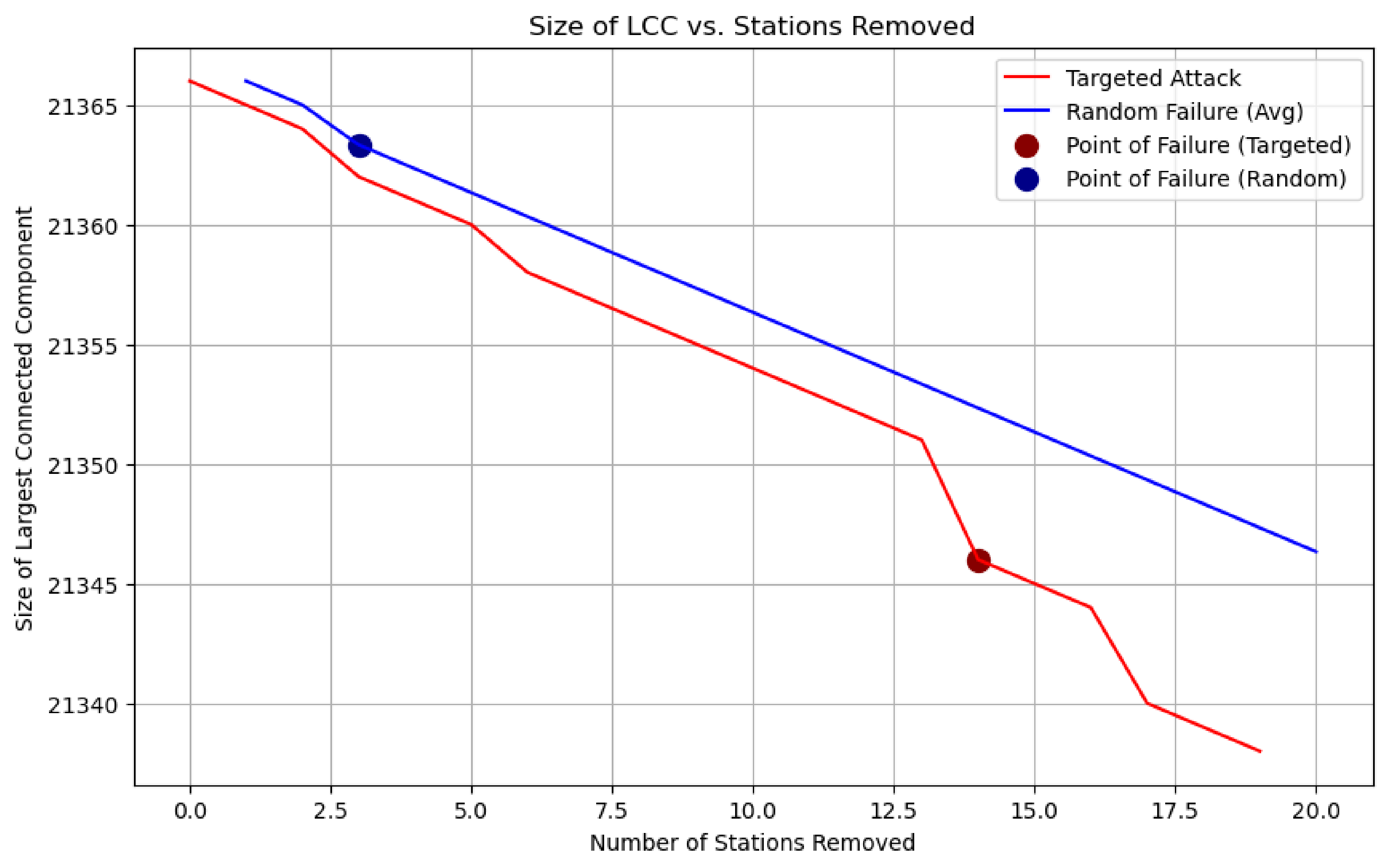
## 2. Calculate <u>LCC size</u> and the <u>APL</u>

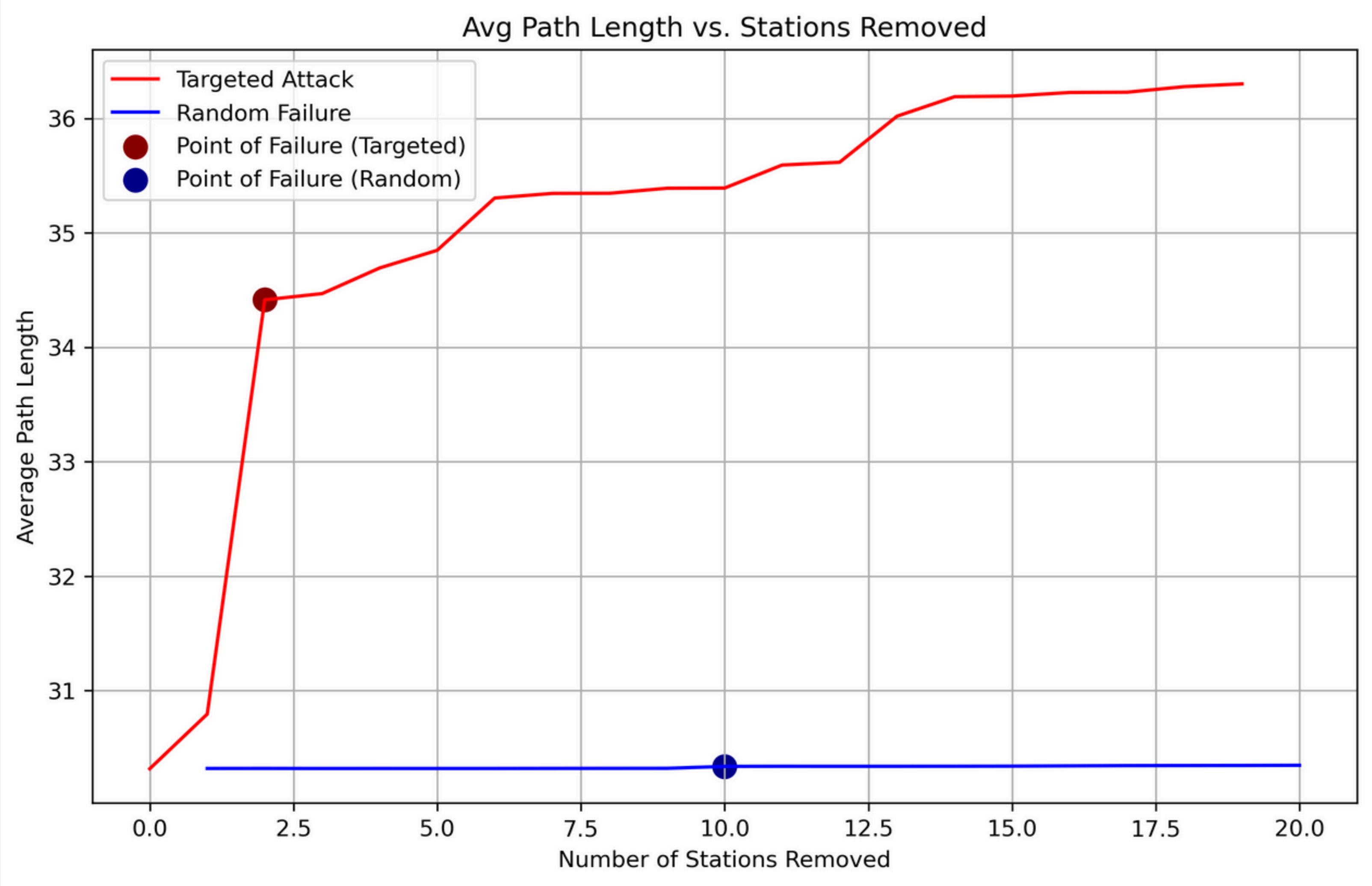
- **LCC Size** (Largest Connected Component):
  - Measures the size of the largest remaining part of the network as stops are removed
- **APL** (Average Path Length):
  - Tracks how efficiently stops remain connected

## 3. Compare results and make conclusions

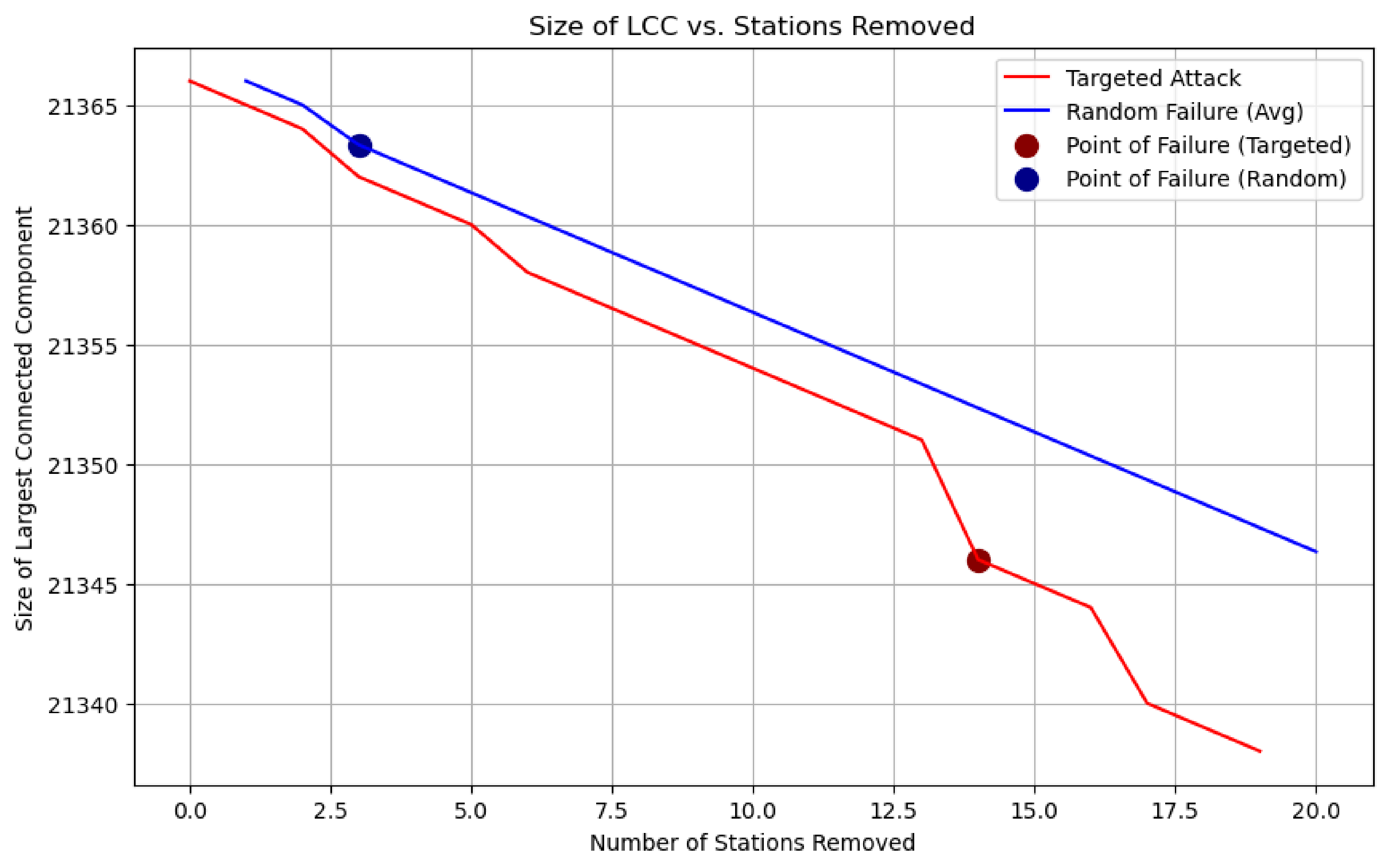
- Identify which network (London vs. LA) is more resilient and why
- Discuss how network design impacts overall robustness

# Analysis – London



# Analysis - London

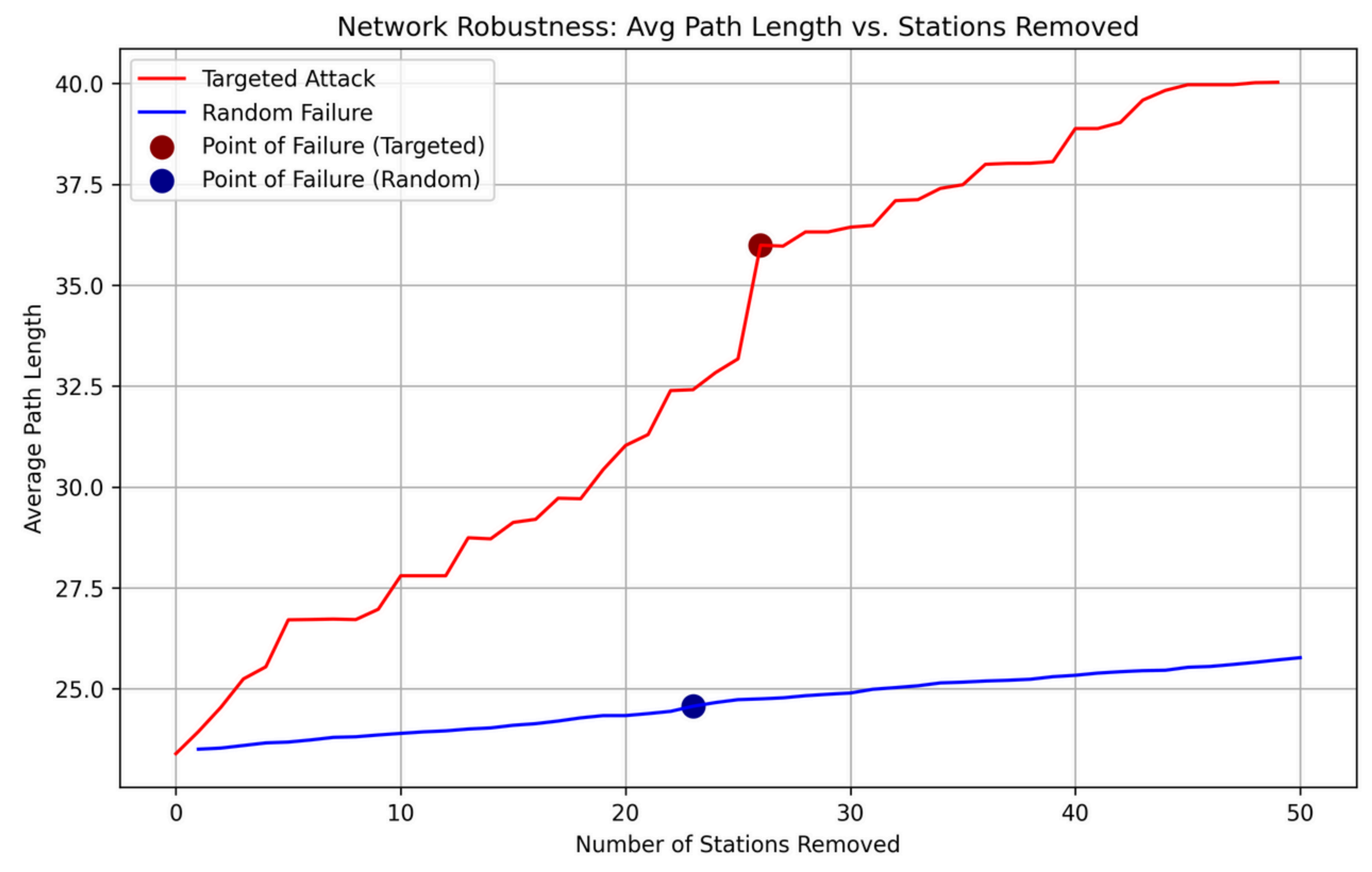



## Size of LCC vs. Stations Removed

- Shows how the LCC shrinks as stops are removed
- **Targeted Attacks:**
  - Targeted removals cause a significant drop earlier
  - Shows the network's vulnerability when critical nodes are removed intentionally
- **Random Failures:**
  - Removing random stops causes a more gradual decrease in LCC size than the targeted attack
  - Due to its definition, point of failure occurs early on as it only experiences a significant change then, but afterwards the decrease is very slow and gradual

**Targeted attacks break apart the network quicker because high-betweenness stops are important for connectivity**

# Analysis - London

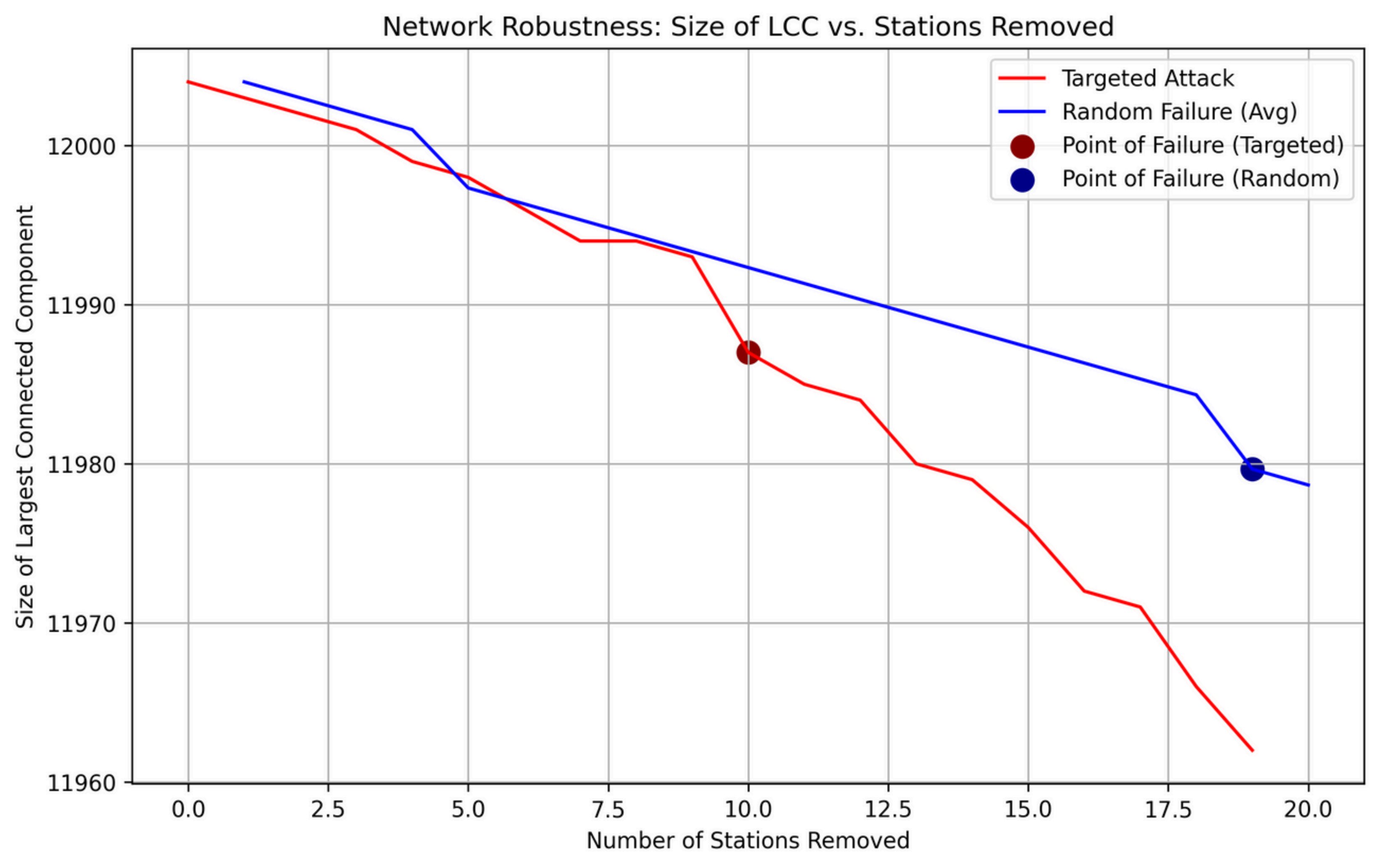


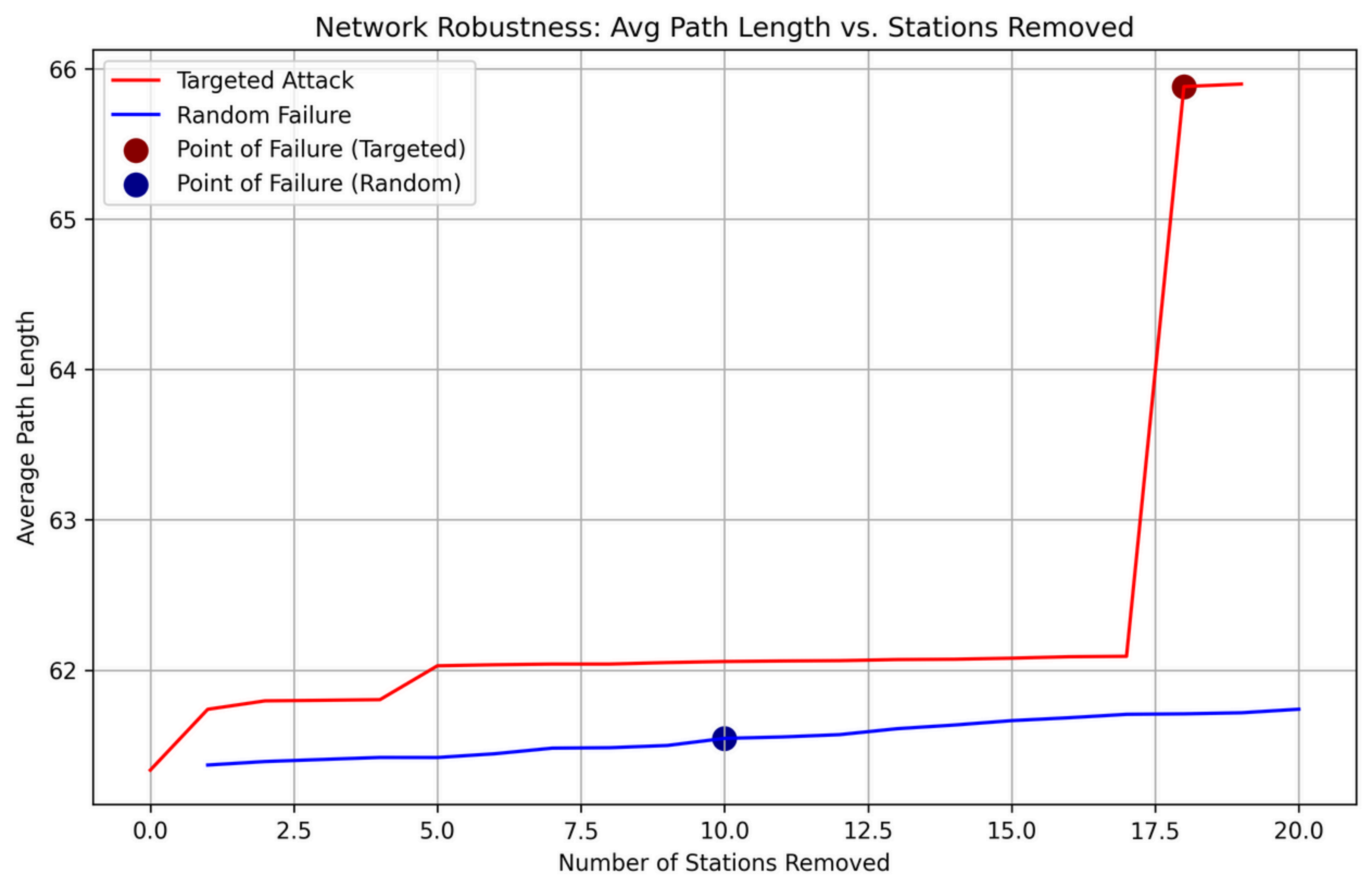

## APL vs. Stations Removed

- Shows how the APL increases as stops are removed
- **Targeted Attacks:**
  - APL increases quickly, then jumps at the point of failure ( ~25 stops removed)
  - Overall, the APL increases quite significantly, showing loss of travel efficiency in the network
- **Random Failures:**
  - APL increases very slowly as random nodes are removed
  - Point of failure shows only a minor disruption
  - Random removals have minimal impact on the APL, which shows resilience to random disruptions

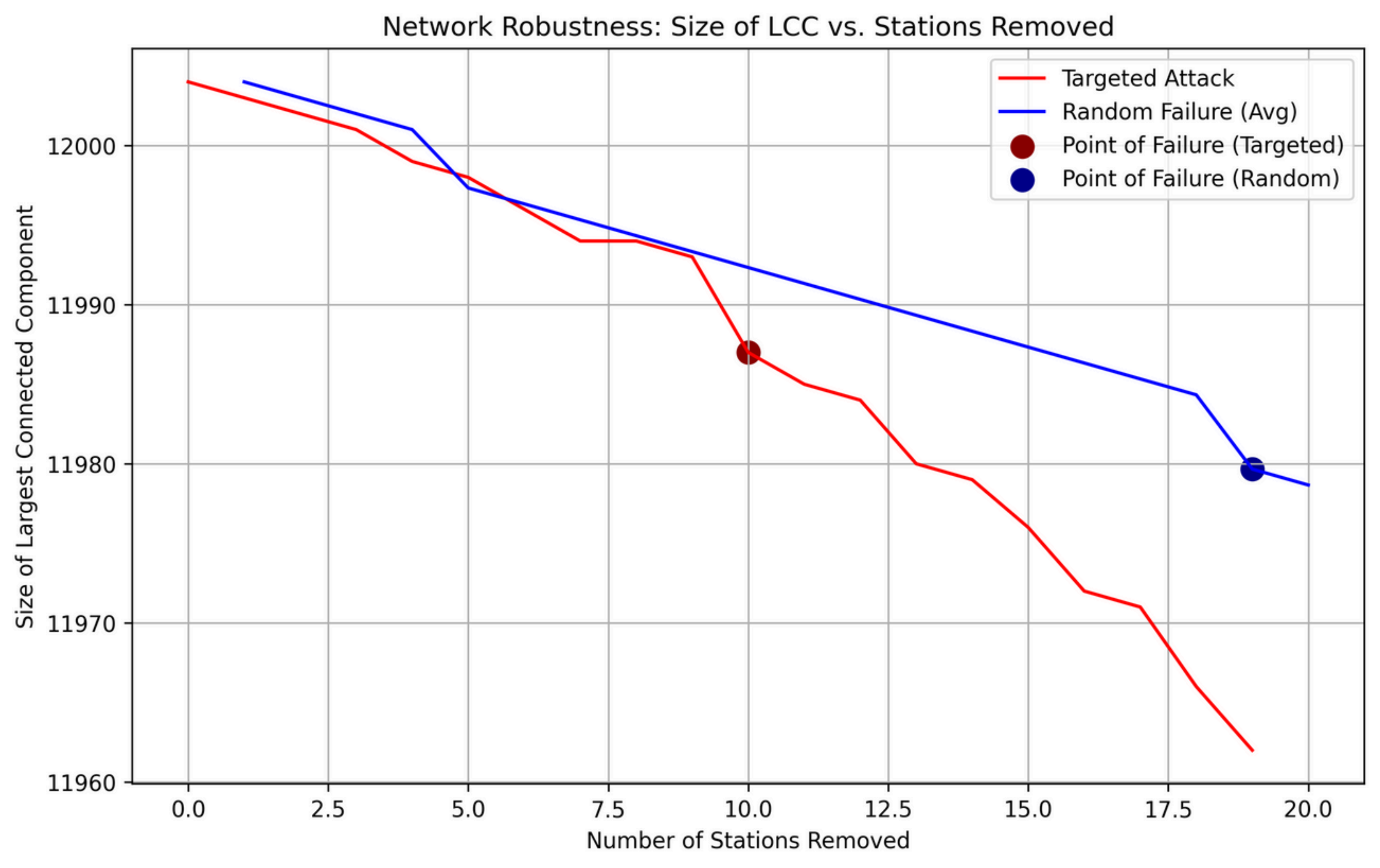
**Targeted attacks cause a loss of efficiency and random failures have a much smaller impact, with only gradual increases in path length**

# Analysis - Los Angeles



# Analysis - Los Angeles

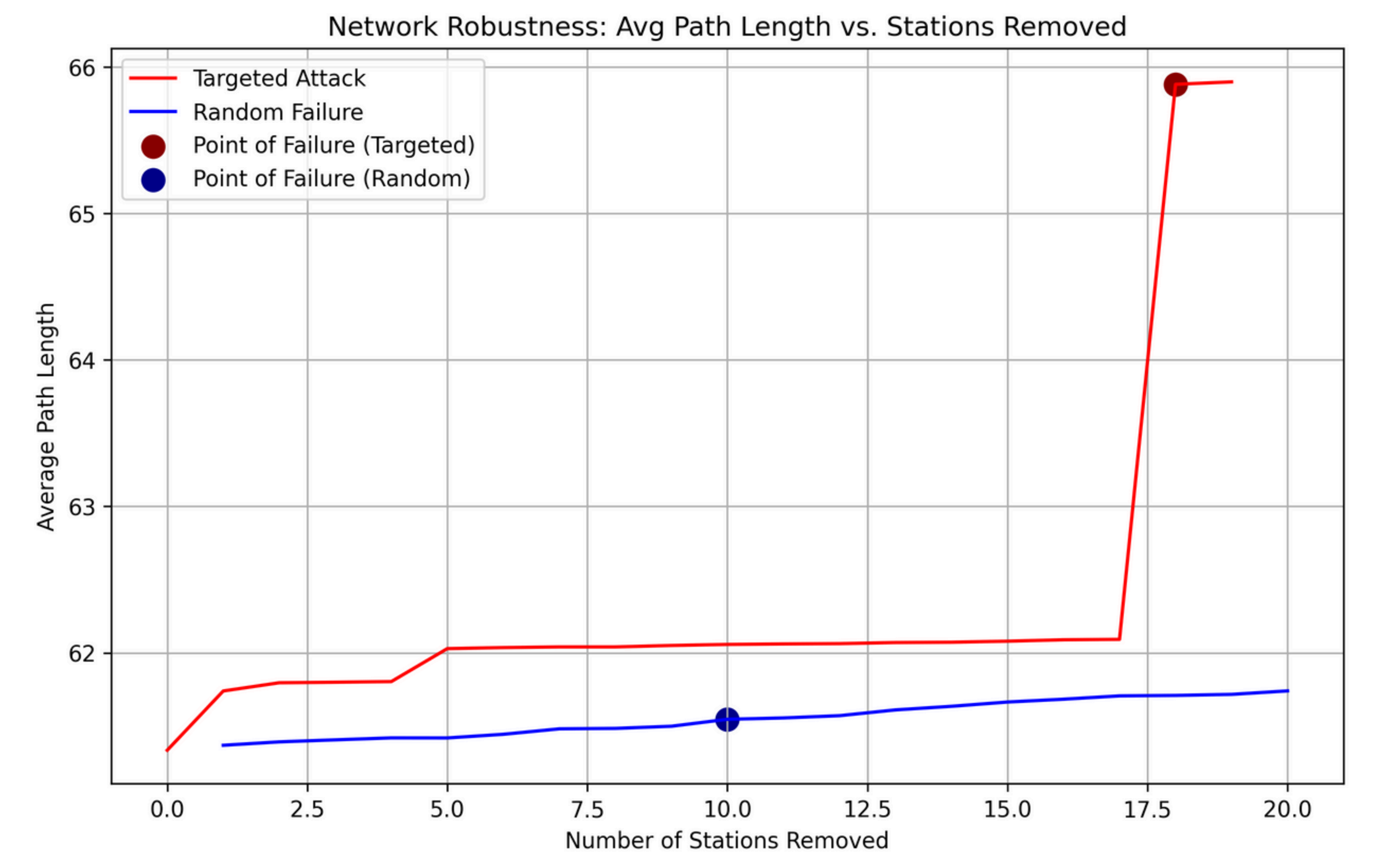



## Size of LCC vs. Stations Removed

- Shows how the LCC decreases as stops are removed
- **Targeted Attacks:**
  - LCC size drops gradually after just a small number of critical stations are removed
  - Point of failure (~10 stations) shows where the network begins to fragment more quickly
- **Random Failures:**
  - LCC size decreases gradually as random stops are removed
  - The point of failure (~18 stations) occurs later

**LA network seems dependent on critical hubs**
**Under targeted attacks, network falls apart quicker. Random failures have a smaller, gradual impact, showing redundancy in the network**

# Analysis - Los Angeles




## APL vs. Stations Removed

- Shows how the APL increases as stops are removed
- **Targeted Attacks:**
  - APL remains relatively stable at first but spikes at the point of failure (~10 stations removed)
  - Network becomes more inefficient when critical stops are removed
- **Random Failures:**
  - APL increases slowly, showing that random node removal does not significantly disrupt the network's efficiency

**Targeted Attacks cause a sharp increase in APL. Random Failures only have a minor effect on APL, suggesting that the network maintains its efficiency well**

# Results

## London

- Overall more resilient to disruptions
- Handles random failures well, showing slow and gradual loss in connectivity and efficiency
- Targeted attacks cause quicker fragmentation and loss in travel efficiency

**London's network is more robust overall, but still relies on critical stops for maintaining good connectivity**

## Los Angeles

- Less resilient, especially under targeted attacks
- Network breaks down quicker when critical stops are removed
- Random failures cause some disruptions, but the network remains less efficient overall

**Los Angeles' network is more fragile under targeted attacks due to its reliance on critical hubs and lack of alternative routes**

# Research Question Answer

- **London is more resilient overall**
  - It handles random failures well, with gradual decreases in connectivity and efficiency
  - Targeted attacks on critical stops cause more and quicker disruption
- **Los Angeles is less resilient under targeted attacks**
  - Network breaks down quicker when important stops are removed
  - Network remains less efficient than London’s network overall

**London’s network has a stronger overall structure, staying mostly connected under disruptions. However, its efficiency drops quicker, as seen in the sharp increase in APL**
**In contrast, Los Angeles fragments faster losing connectivity earlier, but its APL remains relatively stable due to the network’s already lower base efficiency**

# Limitations and Further Research

## Liminations

- **Unweighted Network Assumption**
  - The analysis treats all connections between stops as equal
  - Real-world bus networks have weights like travel time, frequency of buses, or passenger capacity, which can greatly affect resilience
- **Static Network Representation**
  - The network does not account for time-dependent changes like, peak vs. off-peak hours, bus delays, or dynamic disruptions (multiple failures happening sequentially)
- **Lack of Passenger Data**
  - The importance of stops is determined based on centrality measures, but real-world importance depends on passenger flow
- **Computational complexity**
  - APL as a comparative measure does not work best for large, dense networks like London because it's very computationally expensive

## Further Research

- **Weighted Network Analysis**
  - Incorporate edge weights like travel time, frequency, or passenger volume to study how weighted disruptions affect resilience
- **Edge Disruptions**
  - We focused on node (stop) removal, but routes (edges) can also fail due to traffic, construction, or other disruptions
- **Compare more cities**
  - Expand the analysis to include cities with different urban layouts and public transport designs
- **Improving Network Design**
  - Propose ways to make public transport networks more resilient, reduce dependence on important stops and optimize stop placements to improve connectivity and efficiency
- **Choosing metrics**
  - Explore and implement alternative comparative metrics to gain better insights into network performance

# Thank you!

for your time